COPYRIGHT RESERVED BCA(II)—Math (GE – 2)

2025

(Session : 2024-27)

Time : $1\frac{1}{2}$ hours

Full Marks : 40

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

1. Answer any **four** questions of the following : 6×4 = 24

(a) Prove that the inverse of an element in a group is unique.

(b) Prove that the inverse of the product of two elements of a group is the product of the inverse taken in reverse order.

(c) If W_1 and W_2 be two subspaces of a vector space V of finite dimension, then show that dim $(W_1 + W_2)$ = dim W_1 + dim W_2 – dim $(W_1 \cap W_2)$.

(d) Verify union of two subspaces of a vector space is a vector space or not with suitable examples.

(e) Find the adjoint of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & -1 & 3 \end{bmatrix}$$

and verify a (adj. A) = (adj. A) A = |A|I.

(f) Find the rank of the matrix A where

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \\ 3 & 0 & 5 & -10 \end{bmatrix}.$$

Section – B

2. Answer any **four** questions of the following : 4×4 = 16

(a) Prove that intersection of two subgroups of a group is a subgroup.

(b) If R is a ring, then for all a, b, c ∈ R show that :

(i) a.0 = 0.a = 0

(ii) a. (– b) = – (ab) = (– a).b

(c) If F be a field and F′ be a subfield of F, than show that F(F′) is a vector space over F.

(d) If W_1 and W_2 be two subspaces of a vector space V then prove that $W_1 \cap W_2$ is a vector subspace of V.

(e) Solve the following system of equations by matrix method :

$$2x - y - z = 4$$
$$-x + 2y - z = -5$$
$$x - y + 2z = 1$$

(f) Find the eigenvalue of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

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